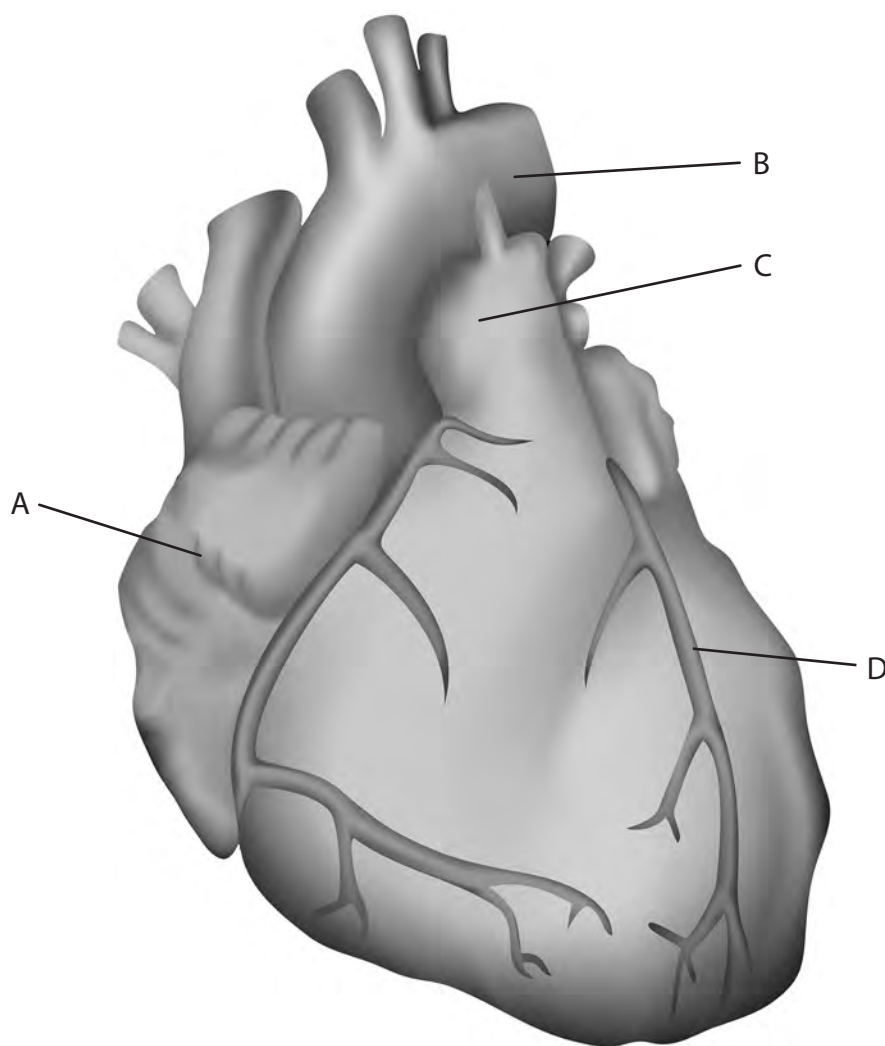


3 The structure of a heart can be investigated by dissection.

(a) The drawing shows an external view of a mammalian heart.



(Source: © VERONIKA ZAKHAROVA / SCIENCE PHOTO LIBRARY)

(i) Name the structures A, B, C and D shown in the diagram.

(2)

A

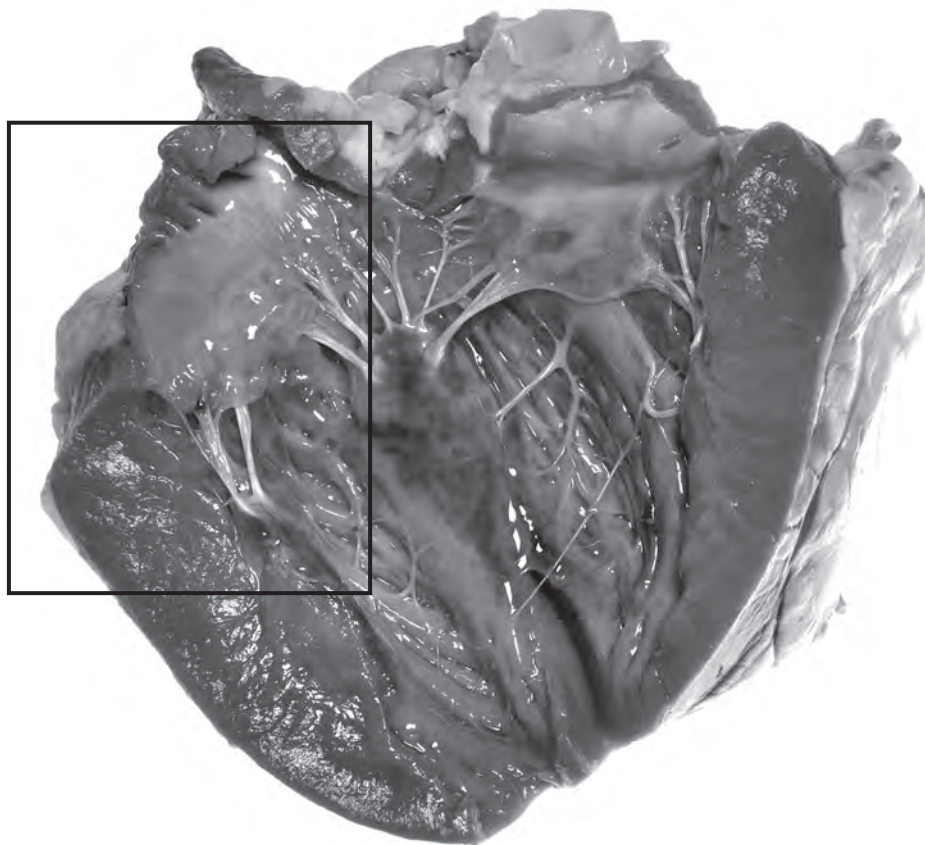
B

C

D



(ii) The photograph shows a heart after it has been dissected.



(Source: © PjrStudio / Alamy Stock Photo)

Draw a plan of the structures within the rectangle shown on the photograph and label two of these structures on your drawing.

(3)



(iii) State the function of the **two** structures labelled on your drawing.

(2)

Label

Function

Label

Function

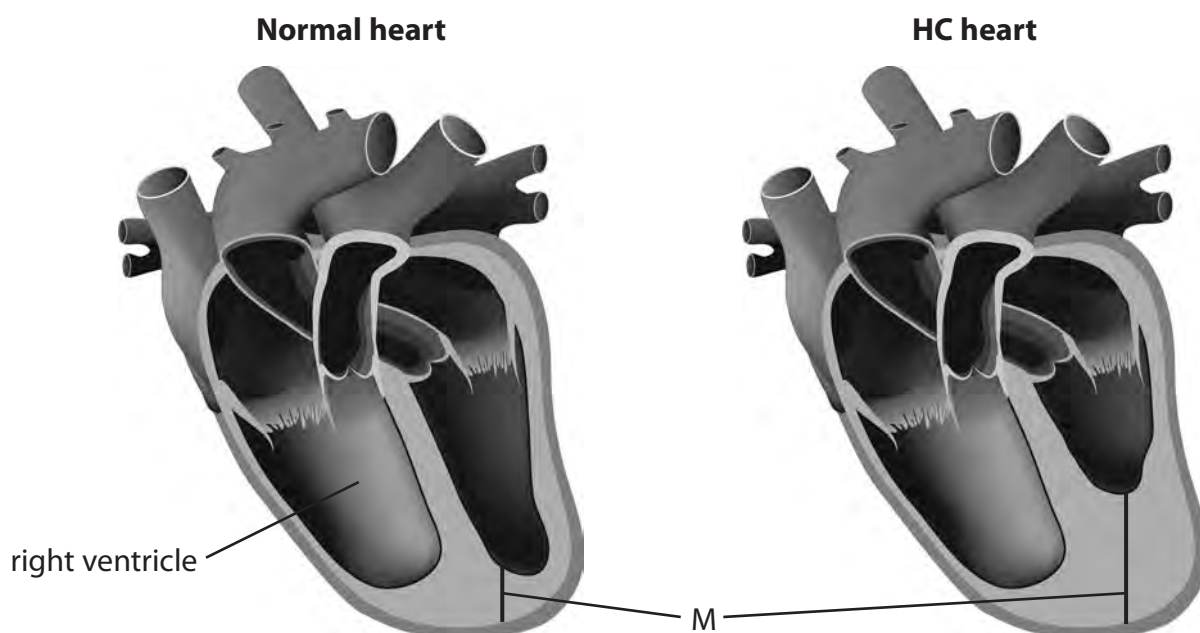
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(b) Some people inherit a type of heart disease called hypertrophic cardiomyopathy (HC).

In HC some of the muscle in the heart becomes thickened, as shown in the diagram.



© designua/123rf

(i) Calculate the percentage increase in thickness of the muscle in the HC heart compared with the normal heart, using the lines labelled M.

Give your answer to two significant figures.

(2)

..... %



(ii) Compare and contrast these two hearts.

(4)

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P 6 6 1 5 5 A 0 1 5 2 0

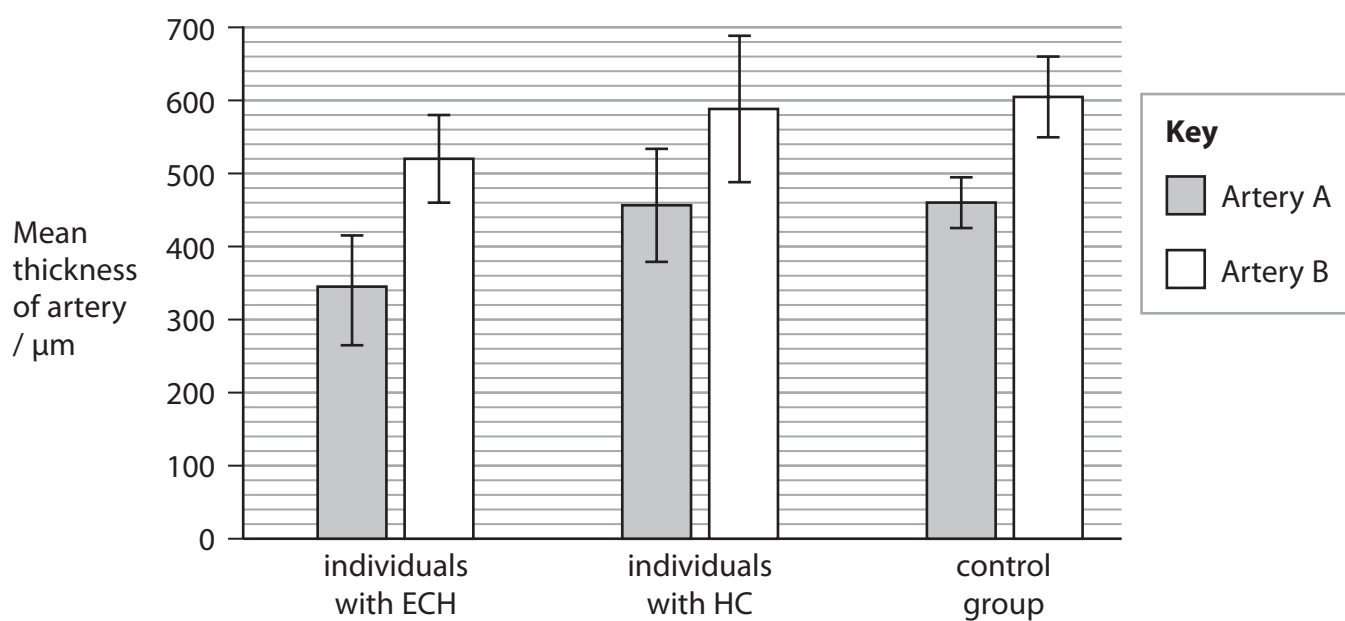
- (iii) Athletes can also develop cardiac hypertrophy as a result of exercise programmes (ECH).

Individuals with HC and individuals with ECH show changes in the thickness of the walls of some of their arteries.

In a study, the thickness of the walls of two of these arteries was measured in three groups:

- individuals with ECH
- individuals with HC
- a control group without ECH or HC.

The graph shows the results of this study.



Draw a table to show the results of this study.

(3)

(iv) Determine the effects of ECH and HC on artery thickness.

(4)

(Total for Question 3 = 20 marks)

TOTAL FOR PAPER = 50 MARKS

